Chapter – 2

Technical Communication

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Background information on TECHNICAL COMMUNICATION

- **Communication:** transfer of information
- **Technical information:** information related to engineering, evaluation, and research and development **(R&D)**
- **Information associated with** design, production, operation, use, and/or maintenance of an equipment, machine, process, or system.
- **Information related to the** use of goods and services.
- **Communication about** technical products and services

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- **Transfer of technical information**
- **Communication which is intended to inform and enable people to use the technical goods and services well.**
- **Communicating about how to use a product and service easily, safely, effectively and efficiently.**
- **The process of creating, designing and delivering information about products for its safe, efficient and effective use. (technical systems, software, services).**

Examples

- **computer manuals (print versions and help screens)**
- **assembly instructions for appliances, equipment, furniture, toys, and games**
- **research articles that present scientific or technological discoveries**
- **magazine articles that explain how to complete a process—preparing yellow chicken curry or a dovetail joint**
- **training films that demonstrate a process**
- **instructions on how to register online or create an online account**
- **wikkis or user groups that exchange and/or continuously update technical information or document processes**

2.1 Defining Technical Communication

- **“Technical communication is a process of managing technical information in ways that allow people to take action”**

  - Richard Johnson-Sheehan

- **“Technical communication is the process of creating, designing and transmitting technical information so that people can understand it easily and safely, effectively and efficiently. Most technical communication is written by people working in or for organizations. Technical communication is reached by people who need to carry out procedures and solve problems.”**

  - Mike Markel

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- **“It is the process of gathering information from experts and presenting to an audience in a clear, easily understandable form”.**

  - Society for Technical Communication (STC)

- **A technical writer’s main task is to convey information to another person in a very clear manner.**
- **Examples: user manuals, press releases, memos, business proposals, product descriptions, specifications, resumes, job applications.**

2.7 Importance of Technical Communication

- Technical communication is a way of facilitating communication about projects and ideas throughout the industry.
- It is important in documenting discoveries, designs, products, and methodologies.
- It helps convey information about the field to non-technical people and external agencies for the purposes of support, administration and funding.
- It contributes to usability of designs, and it provides information for end users so they can understand and enjoy the resulting products and processes.

1. Personal value

2. Workplace communication

3. Complex environments

Personal value

- The new media has penetrated our personal lives. In the past, new technology was limited to work environment or businesses.
- Today, our lives are deeply intertwined with computers, smartphones, and electronic devices. To navigate them effectively, mastering basic personal communication like emailing, texting, browsing, and social media interaction demands solid technical communication skills.

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- Technical communication in the form of proper language use and effective interpersonal skills is equally important.
- It is needed for thinking clearly, organizing information logically and presenting it effectively.
- There lies its importance in easy, safe, effective and efficient use of technical product and services.

Workplace Communication

- Every organization, small or big, depends on communicating to get things done.
- Such communication in small organizations may involve interpersonal exchanges and some written materials, but in large organizations the volume of formal , oral as well as written communication is bigger, requiring the skills of a technical communicator.

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- Everyone within an organization needs some technical communication skills: managers, supervisors, workers.
- Technical communication goes hand in hand with product development.
- The developer develops a product or service, and the technical communicator writes documentation on it.
- Inefficiencies or errors in communication will cost a company money, resources, time and ultimately its image.

Complex Environment

- Organizations are becoming complex in size, structures and areas of work because of technological penetration, global business advancement and diversifying workforce.
- It requires well-coordinated communication process both internal and external.
- More specialized skills in research, teamwork, negotiation, deliberation and interpersonal as well as translation, and multicultural communication are in demand.

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- Without the technical communication skill the product developers will not be able to explain how to use the product or how it may benefit the users.
- Globalization of trade and business continues to expand the scope for technical communication
- Outsourcing has made technical communication even more relevant to developing markets like Nepal.
- Technical communication in advanced economies is a thoughtful business.
- Today’s complex technological environments also call for multipurpose technical communication skills.

2.3 FORMS OF TECHNICAL COMMUNICATION

- **BASICALLY:**

- ORAL AND WRITTEN

- **OTHER FORMS**

-HIGH-TECH AND LOW –TECH

-INTERNAL AND EXTERNAL

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- **ORAL-** sense of immediacy, personal and informal touch, flexibility, high level of transparency, spontaneous feedback, receptivity, interactivity,, time saving, effective in resolving conflict

Disadvantages- less authentic, transcribing can be costly, messages can be long or distorted, requiring great receptivity and attentiveness

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- **WRITTEN-** good for emphasis, evidence, records and reference, more organized, precise, explicit and permanent, leaves no room for misunderstanding, allows for clarification or verification, sounds formal and authoritative, time saving, reliable for lengthy content or data.

Disadvantages- costly in production and delivery, involves paperwork and burden, requires skilled manpower to write, impersonal, response is not spontaneous

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- High-tech- software interface, computer applications, medical procedures etc.
- Low-tech- memos, reports, newsletters, websites etc.
- Internal – reports, proposals, presentations, memos ,minutes, newsletters etc.
- External – brochures, flyers, public announcements, product manuals, releases, posters etc.

2.5 TYPES OF TECHNICAL COMMUNICATION

- Mail/ correspondence
- Documentation/ Records
- Forms/ Receipts
- Planning
- Procedures/ descriptions
- Instructions/ manuals/ guides
- Proposals/ agreements/ policies
- Oral presentations/ group work
- Reports

Types of Technical Communication

a. Mail/Correspondence

Business: sales letter, quotations letter, official: circular letter

Professional: Acceptance letter; application letter, appointment letter

b. Documentation/Records : Meeting minutes, class notes, speech or presentation

c. Forms/Receipts: work order form, Tax form, application form

d. Planning: meeting agenda, action plan.

e. Procedure/Descriptions: procedures for activities, operation guides, catalogues.

f. Instructions/Manuals/guides: instruction writing, user manual

g. Proposals/Agreements/Policies: call for proposal , Terms of Reference, budget estimate, Treaties

h. oral/presentation/group work: oral reports, group discussion, webinar

i. Reports: Progress Report, lab Report, survey Report

j. Evaluation : Audit Report, feasibility Report, Review

k. Promotion/Outreach/ Information: leaflet, pamphlet, banner, hoarding board, poster, notice

l. Design and Technology: Email, CDs Websites, programming.

2.2 Evolution of Technical Communication

- The **root** of technical communication goes back to **classical antiquity**.
- It can be **traced** to ancient **Greece and Rome** as well as to **Sumeria, and ancient Egypt**.
- **Handed down orally first**, such communication **contained descriptions of scientific and astronomical observations.**
- The works of the writers like **Aristotle** can be taken as **the earliest form of technical writing.**

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- **During or around Renaissance period (1400-1600),** Copernicus, Geoffrey Chaucer, Leonardo da Vinci and Newton wrote technical details and drawings on the use and operation of their various inventions.
- **The earliest document- Chaucer's 'Treatise Astrolabe'** (medieval manual on astrolabe- *a scientific instrument used to measure the position of heavenly body*)
- The field of technical communication grew during industrial period.
- **Samuel Chandler Earle,** who wrote the book, **The Theory and Practice of Technical Writing** (1911), is generally considered to be **the father of technical writing**.

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- During 20th century, the need for technical writing skyrocketed.
- **During the First World War**,, western governments spent large amount of money to write manuals and instructions on the use of weapons and defense products.
- **Following the war,** large telecom and car manufacturers and other companies created technical communication departments within their organizations.

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- In 1949, Joseph D. Chapline **wrote the first computer related technical manual**. He also wrote a brief pamphlet, ***Technical Writing***.
- **In 1953**, it came together to form professional associations in the USA.
- It gained further pace with the growth and expansion of technical industries, particularly electronics and computers, as various government legislations made it compulsory for organizations to produce certain technical documents as part of ensuring standards to meet public obligations (Eg. ISO 9000 in the US).

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- Today, the field has expanded around the world, with most of the technical communication taking place in countries with an edge in hardware, software, e-commerce etc.
- Experts have estimated that India has possibly the largest number of technical writers outside North America.

In Nepal

- The scope of technical communication has been widened in Nepal due to steady growth of technology companies, production of websites, the gradual association of the private sector and NGOs and modernization of various professions.
- The traces of technical communication practices can be gathered from scriptures, oral traditions and more recently from social histories.

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- ***Ayurved,*** the medical treatise, from more than **5000 years ago**, is full of instructional technical information on healing practices.
- ***Manusmriti,*** the law of Manu, looks equally technical.
- During the Golden Age, **Arniko,** the master constructor, must have written or at least sketched out some technical documents on pagodas and chaityas.
- In villages, across the countries, **lekhandas,** for many generations, helped crack the legalese for the masses.
- **Jyotishis and Vaidhyas** are also the technical communicators.

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- **Technical content exists in abundance today, but apparently, Nepal has few effective technical communicators to translate them for non-specialists.**
- **We need to prepare good number of technical communicators to facilitate the people about technical products and services.**

Technical Communication Competencies

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4.1 Defining Competency

4.2 Attitude, Values and Personality

4.3 Core Competencies in Technical Communication

4.4 Skill Sets for Technical Writers

4.5 Assessing Competencies

4.6 Occupational Outlook

Meaning of competency

- On the basic level, competencies are certain abilities coupled with knowledge and experience.
- a combination of knowledge, skills, abilities, and behaviors that enable individual to perform specific tasks/ roles effectively.
- Skills grow and develop with understanding, opportunities to articulate and demonstrate the acquired knowledge and put it into practice.
- Competency pyramid shows, competence is a specific knowledge transformed into action to address a specific problem.

Competency Pyramid

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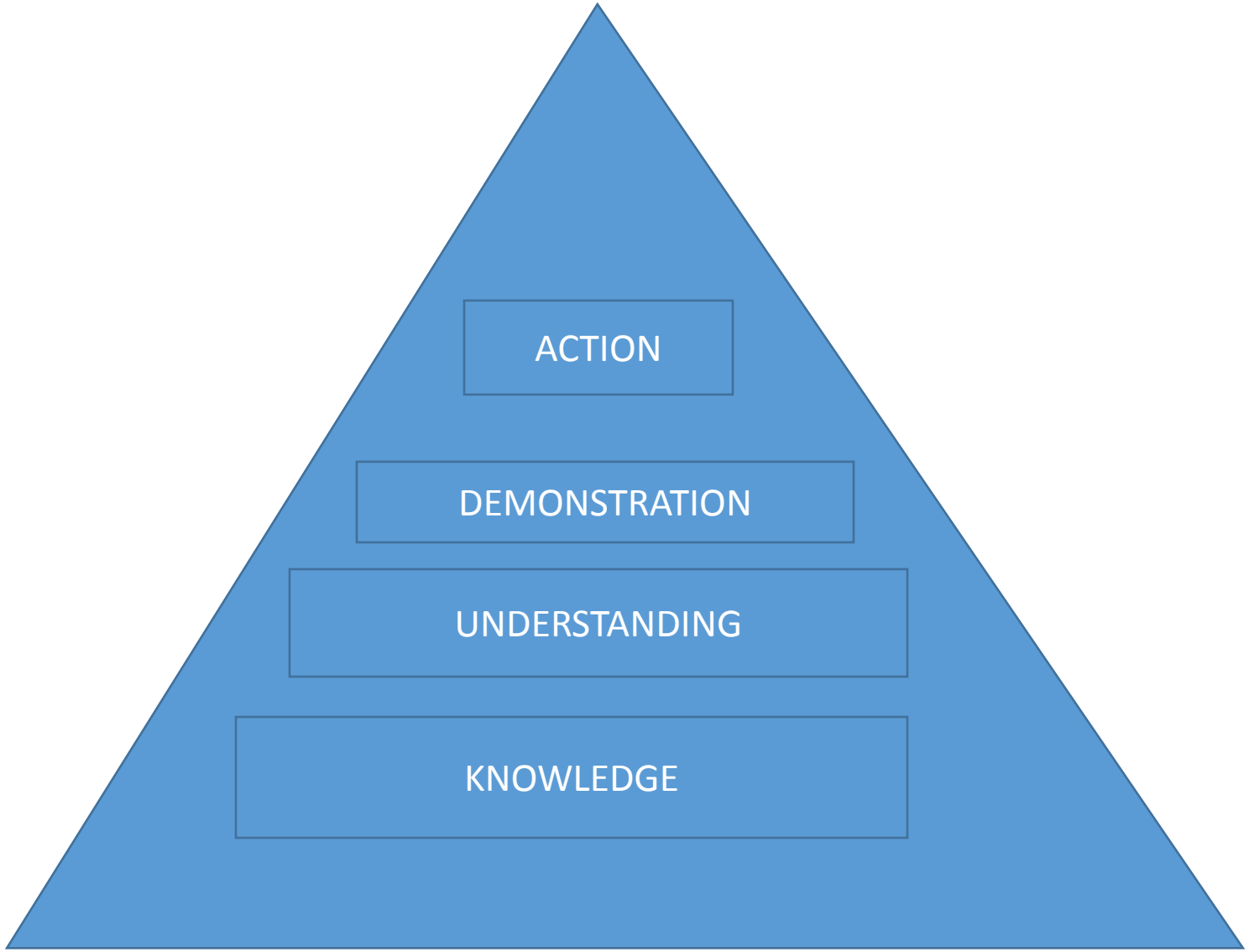

4.1 Defining Competency

- "Sufficiency of qualification; capacity to deal adequately with a subject."
- The root word here is 'competence' meaning ' ability to do something successfully or efficiently.'
- Michael Eraut (1994) explains that competence refers to specific capabilities, meeting people's expectations.
- Competency is either performance capability or a piece of useful knowledge or skill.
- Technical communication competency- the ability to communicate technical information competently.

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- **Prerez and Temporada (2010) describe competencies as ‘ the underlying and broad-based characteristics and grouping of knowledge, skills, behaviors, attributes, and values in an individual that cause or predict superior performance.’**
- **Conclusion-** ***competencies are the abilities to perform specific real-world tasks.***
- ***They are primarily gained through doing, learning, training etc.***
- ***They are important in human resource management to assess the skills needed for specific jobs and their performance standards.***
- ***The present need is competent technical communicators who can contribute to both internal and external operations of the business organizations.***

Core Competencies in Technical Communication

- Each task involves specific, targeted skills and knowledge.
- In the business or organizational context, the required sets of skills and knowledge are often referred to as “core competencies.”
- Society for technical communication identified the following competencies
- **Advocacy:** Ability and willingness to be an advocate/activist for the user.
- **Design:** knowledge of information design, presentation of data, language conventions, communication principles and theory.
- **Execution:** Ability and willingness to apply information design, language, and communication models, theories, rules, and standards.
- **Innovation:** Ability and Willingness to be open to new ideas without sacrificing usability or accuracy.

- **Use of media:** Ability and willingness to understand the requirements and uses of different media and to apply them appropriately**.**
- **Research skills:** Ability and willingness to gather relevant and accurate information and analyze it for appropriateness.
- **Use of supporting tools:** Ability and willingness to use appropriate supporting tools, including computer application software etc.
- **Usability_** Understanding of usability, skill in user and task analysis, and the ability and willingness to provide value to users of the information.

George Hayhoe identified four areas of core competencies.

- **Creating and managing knowledge:** Technical communicators must be more than "packagers of information for the technically unskilled." Instead, they must be masters of the technical domains they write about.
- **Designing information that readers need:** Technical communicators must "provide information that users need by carefully selecting the right mix of content and then developing, arranging and presenting it effectively for the audience.
- **Communicating fluently in various media:** Technical communicators must be fluent not only in writing but in Video, animation, photography, or illustration.
- **Being part of a learning community:** Technical Communicators associate with like-minded academic or professional groups.
- STC enables technical communicators to acquire and maintain essential skills and knowledge in the profession.

- David Dayton, Kenneth T. Rainey and Roy K Turner identified core competencies for technical communicators are:
- **Collaborative skills:** Skill in collaborating with both subject-matter experts and coworkers.
- **Writing skills:** Ability to write clearly for specific audiences directed by clearly defined purposes**.**
- **Technical skills:** Ability to assess and learn to use technologies.
- **Self–activation/evaluation skills:** to evaluate one’s own work and the work of others.

Technical Writer: Which Skills Are Important?

- Technical writer requires a mastery of a certain set of skills.
- These skills involve primarily writing and illustration skills, as large manuals for print publication, standard in the profession.
- **Writing Skills:** Needs to write in a clear and concise manner and to be able to convey information appropriately for a variety of audiences.
- **Technical skills:** The technical skill depends greatly on the subject matter, product or service that requires documentation.
- Pharmaceuticals and other manufacturing industries have specific requirements of transmitting technical knowledge
- **Tools skills:** a technical writer needs to know about computer systems since they are used to produce documentation in a variety of formats.
- **Specific tool knowledge**, such as Adobe FrameMaker, MS Word, MadCap Flare, RoboHelp, and even PageMaker and Quark really depends on the tools the organization has come to rely on in order to produce its technical documentation.

- **Interviewing and listening skills:** Technical writers need to know how to ask questions.
- Strong listening skills will be required to capture the necessary information, and need to consider the varying personalities and preferences of the people to know how best to approach them as well as which follow-up questions need to be answered.
- **Design skills:** An appreciation for the visual can be an important part of the skill set of a technical writer.
- To a growing extent the technical writer needs an appreciation for graphics and formatting as well as illustration skills.
- **Usability and testing skills:** The technical writer knows that validation of the documentation is important-the confirmation that the product works the way it is documented to work.

4.2 Attitude, Values and Personality

Attitude

- **Attitude is a state of mind, feelings, or beliefs directed at a specific person, institution, object, or policy.**
- **It is evaluative in nature.**
- **It is a mental outlook, frame of mind, point of view, perspective, and way of looking at things.**
- **It may be positive, negative, or neutral.**
- **It is a settled way of thinking or feeling about something.**

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- ***In psychology, attitude is a psychological construct, a mental and emotional entity that inheres in, or characterizes a person.***
- ***It is complex and acquired through experiences.***
- ***It is an individual's predisposed (inclined) state of mind regarding a value and it is precipitated (caused) through a responsive expression toward a person, place, thing, or event (the attitude object) which in turn influences the individual's thought and action.***

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- **Attitudes are not behaviors, although they may influence others.**
- **It can be flexible and may not be as stable as competencies.**
- **They may change over time.**

Types-

- **Cognitive (how an individual perceives facts)**
- **Affective (feelings and emotions)**
- **Evaluative (positive or negative towards an object or task)**
- **Behavioral (intentions or desires regarding the performance of a task)**

Values

- **Values are beliefs or principles that guide behavior and decision-making.**
- **They have deeply held convictions about what is important and meaningful in life.**
- **Values can be individual or collective and are shaped by a range of factors, such as family, culture, religion, education, and personal experiences.**
- **Influence our judgments, preferences, and choices in life.**
- **Values shape our decisions.**
- **Compared to attitudes, values are stable beliefs, consistent across time and context.**

Types

- **Personal values** (personal principles of everyday life)
- **Work values** (standards of work outcome or job satisfaction)
- **Social values** (norms, customs, cultures of people or communities, organizations)
- **Life values** (ultimate aspirations/goals)

Personality

- Personality refers to a unique set of psychological traits, patterns, and characteristics that define an individual's behavior, thoughts, and emotions.
- A set of qualities that makes a person distinct from others.
- A person’s patterns of traits are characteristics such as being helpful, sociable, and independent.
- These patterns are generally distinguishable, separating one person from another.
- They are not necessarily evaluative.
- They are innate, not necessarily learned on the job.

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Icek Ajzen (2005) identifies nine personality traits.

- **Dominance**
- **Sociability**
- **Independence**
- **Conscientiousness/ Careful**
- **Hostility/unfriendly or aggressive**
- **Helpfulness**
- **Self-esteem**
- **Emotional stability**
- **Ambitiousness**

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- **Some have made the distinction between introvert and extrovert personality, or stable and unstable personality.**
- **For others personalities are realistic, investigative, artistic, social, enterprising, conventional, etc.**

THANK YOU